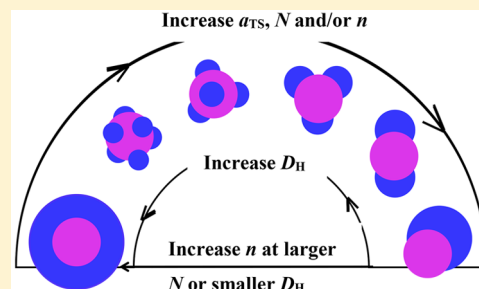


# Simulating Surface Patterning of Nanoparticles by Polymers via Dissipative Particle Dynamics Method

Minqing Gong,<sup>†</sup> Qiuyan Yu,<sup>†</sup> Chenglin Wang,<sup>†</sup> and Rong Wang<sup>\*†</sup>

Department of Polymer Science and Engineering, School of Chemistry and Chemical Engineering, Key Laboratory of High Performance Polymer Material and Technology of Ministry of Education, State Key Laboratory of Coordination Chemistry and Collaborative Innovation Center of Chemistry for Life Sciences, Nanjing University, No.163, Xianlin Road, Nanjing 210023, China

**ABSTRACT:** Patchy particles are often referred to colloidal particles with physically or chemically patterned surfaces. We investigated the patterning of nanoparticle grafted by polymers, mainly consisting of patchy structures with different numbers of patches ( $N_{\text{patch}}$ ) and core-shell structure using the dissipative particle dynamics (DPD) method in good or poor solvents based on the experiment research. Poor solvent, large nanoparticle, proper grafting density and medium polymer length contribute to the formation of patchy structure. We introduce the effective volume fraction as an indicator to distinguish the patchy structure from core-shell structure. The reversible transition between core-shell (in a good solvent) and patchy structure (in a poor solvent) and the dependency relationship between the nanoparticle diameter and grafting density in experiment are verified. Our results pave the way for preparing the colloids with well-defined patches. The anisotropic patchy particles can self-assemble into elaborate superstructures, which are potential blocking materials for drug delivery, sensors, and electronics.



## INTRODUCTION

Patchy particles referring to colloidal particles with physically or chemically patterned surfaces<sup>1–5</sup> can self-assemble into highly ordered, defect-free structures, like colloidal analogues of atoms and molecules<sup>6</sup> because of anisotropic interactions. The elaborated structures of patchy particles are promising building blocks for drug delivery, sensors and electronics.<sup>7–10</sup> In recent years, there have been a lot of researches conducted on the synthesis and theoretical simulation of patchy particles.<sup>1,11–23</sup> Methods drawn from chemistry, physics and biology provide a powerful arsenal for synthesis of patchy particles.<sup>8</sup> Soft colloidal particles obtained by using various block copolymers<sup>2</sup> and polymer blends<sup>24</sup> can self-assemble into diversified patch types. Li et al.<sup>24</sup> fabricated patchy nanoparticles via the assembly of binary polymer blends under the condition of a rapid solvent exchange by applying molecular dynamics. Patchy particles with different numbers of patches, core-shell, Janus and ribbon structures were observed. The results show that patchy particles form due to glassy component, which kinetically traps particle morphology along the path into the equilibrium structures. Patch diffusion and coalescence occur over time, but vitrification of glassy component slow down these processes. The number of patches is determined by the size and the composition of the soft nanoparticles, both of which can be readily controlled by the process parameters specific to the Flash Nano-Precipitation (FNP) process.<sup>25,26</sup> The results are consistent with the experimental results obtained by Chris Sosa et al.<sup>27,28</sup> Ye et al.<sup>2</sup> simply added the poor solvent water into the dimethylformamide (DMF) solution which dissolves polystyrene-*block*-poly(4-vinylpyridine) (PS-*b*-P4VP) and they

gained patchy particles with PS forming soft cores and P4VP forming patches. With the increasing of water, the swelling PS core is driven to form condense core, while the dissolved block P4VP is driven to form patches. Besides, they adopted dissipative particle dynamics (DPD) to simulate the experimental process, and found that the number of patches ranged from 2 to 6. Both of the simulation and experimental results show that the number of patches greatly depends on the composition of the copolymer, and the number of patches increases with the volume fraction of patch component decreasing. Choueiri et al.<sup>29</sup> researched on gold nanoparticle tethered with PS chains, and explored how the chain number, chain length of PS and the diameter of nanoparticle influence the morphology of PS in detail. In addition, they also explored the self-assembly behavior of nanoparticles with different shapes and types decorated by polymers. With the addition of water, the quality of solvent changes from good to poor, which drives uniformly thick polymer layer to form core-shell or break up into well-defined patches due to the interaction between polymers, polymer grafting constraints and the reduction of interfacial free energy. The phase diagram for the diameter of nanoparticle vs. the grafting density of polymer chains matches well with the theoretical phase diagram.<sup>30,31</sup> They concluded that when grafting density is fixed, the number of patches is determined by the ratio of root-mean-square end-to-end of polymers to diameter of nanoparticle.

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We adopted dissipative particle dynamics (DPD) method to investigate the patterning of nanoparticle grafted by polymers, mainly consisting of patchy structures with different numbers of patches ( $N_{\text{patch}}$ ) and core-shell structure using the dissipative particle dynamics (DPD) method in good or poor solvents based on the experiment research.<sup>29</sup> By changing interaction parameter between polymer and solvent, the length and number of polymers coated on the nanoparticle, and the size of nanoparticle, we obtained patchy structures with different number of patches and core-shell structure, constructed several phase diagrams and analyzed how they influenced the number of patches. We observed that poor solvent, larger size nanoparticle, proper number of polymers, and length are favorable to the formation of patchy particles.

## MODEL AND SIMULATION METHOD

We use the DPD method introduced by Hoogerbrugge and Koelman in 1992<sup>32</sup> and successfully applied by Groot and Warren in 1997.<sup>33</sup> It is a fruitful modeling technique for discovering the mesoscale hydrodynamic phenomena accompanied by thermal fluctuation.<sup>34</sup> It is possible for DPD to deal with the complex fluid systems on the level of a coarse-grained model involving long time and space scales.<sup>35</sup> In the DPD simulation, a coarse-grained DPD particle represents a cluster of atoms or molecules.<sup>32,36</sup> The time evolution of the simulation system obeys the Newton's equation of motion:<sup>33</sup>

$$\frac{d\vec{r}_i}{dt} = \vec{v}_i, \quad \frac{d\vec{v}_i}{dt} = \frac{\vec{f}_i}{m_i} \quad (1)$$

where  $\vec{r}_i$ ,  $\vec{v}_i$ ,  $m_i$ , and  $\vec{f}_i$  denote the position, velocity, mass of the  $i$  particle, and the force acting on it, respectively.  $\vec{f}_i$ , the total force acting on particle  $i$ , is given by the following equation:

$$\vec{f}_i = \sum_{j \neq i} (\vec{F}_{ij}^C + \vec{F}_{ij}^D + \vec{F}_{ij}^R) \quad (2)$$

Where the three pairwise interactions  $\vec{F}_{ij}^C$ ,  $\vec{F}_{ij}^D$ ,  $\vec{F}_{ij}^R$  represent conservative force, dissipative force and random force, respectively. They only work on particles within cutoff radius  $r_C$ , beyond which the force is neglected. The conservative force as a kind of soft-repulsive interaction acting along the centers of two particles, is described as

$$\vec{F}_{ij}^C = a_{ij}\omega(r_{ij})\hat{r}_{ij} \quad (3)$$

where  $a_{ij}$  shows the interaction parameter between particles  $i$  and  $j$ . The  $r$ -dependent weight function  $\omega(r_{ij})$  provides the range of interaction for DPD particles with a commonly used choice:  $\omega(r_{ij}) = 1 - r_{ij}/r_C$  ( $r_{ij} \leq r_C$ ), and  $\omega(r_{ij}) = 0$  ( $r_{ij} > r_C$ ).  $r_{ij} = |\vec{r}_{ij}|$ ,  $\hat{r} = \vec{r}_{ij}/r_{ij}$ , and  $\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$ . The dissipative force is defined as

$$\vec{F}_{ij}^D = -\gamma\omega^D(r_{ij})(\hat{r}_{ij} \cdot \vec{v}_{ij})\hat{r}_{ij} \quad (4)$$

where  $\gamma$  is the friction coefficient controlling the magnitude of the dissipative force. It is proportional to the relative velocity,  $\vec{v}_{ij} = \vec{v}_i - \vec{v}_j$ . The random force serves as a heat source to equilibrate the thermal motion of unresolved scales, given as

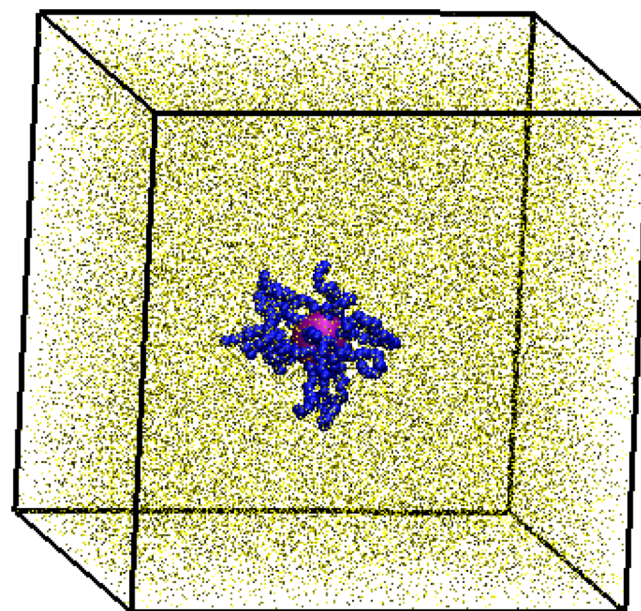
$$\vec{F}_{ij}^R = \sigma\omega^R(r_{ij})\theta_{ij}\hat{r}_{ij} \quad (5)$$

where  $\sigma$  is the noise amplitude governing the intensity of the random force and  $\theta_{ij}(t)$  is a randomly fluctuating variable with Gaussian statistics:  $\langle\theta_{ij}(t)\rangle = 0$  and  $\langle\theta_{ij}(t)\theta_{kl}(t')\rangle = (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\delta(t - t')$ . Español and Warren<sup>36</sup> showed that the relationship between two weight functions  $\omega^D(r_{ij})$  and  $\omega^R(r_{ij})$  is

$$\omega^D(r_{ij}) = [\omega^R(r_{ij})]^2 \quad (6)$$

As a simple choice, we set  $\omega^R(r_{ij}) = \omega(r_{ij})$ , that is,  $\omega^R(r_{ij})$  is the same function as in the conservative force and  $\sigma^2 = 2\gamma k_B T$ . The combined effect of the dissipative and random force amounts to that of a thermostat.

We constructed a coarse-grained model (Figure 1) based on Au nanoparticle with  $n$  tethered polystyrene chains (AuNP-



**Figure 1.** Model of nanoparticle tethered with polymer chains  $T_n$ -H in different solvents.

PS), which comes from experiments conducted by Choueiri et al.<sup>29</sup> We denote the model as  $T_n$ -H, where T represents the polymer chains in blue with length  $N$ , and H represents Au nanoparticle in pink.  $n$  is the number of polymer chains tethered to the nanoparticle. Besides, the solvent is denoted as S in yellow. The diameter of nanoparticle is denoted by  $D_H$ . The size of nanoparticle is alterable and larger than that of polymer beads.

We use the finitely extensible nonlinear elastic (FENE) potential to bind neighboring particles:<sup>37</sup>

$$V_{\text{FENE}}(r_{ij}) = \begin{cases} -\frac{1}{2}kR_0^2 \ln\left[1 - \left(\frac{r_{ij}}{R_0}\right)^2\right], & r_{ij} < R_0 \\ \infty & r_{ij} \geq R_0 \end{cases} \quad (7)$$

where  $k$  is FENE spring constant and  $R_0$  is chosen as  $1.5r_C$ . Simulations are performed in cubic box  $30^3$ . All polymer and solvent beads have the same mass as  $m = 1$ , and the  $k_B T$  is the unit of energy. Here,  $k_B$  is the Boltzmann constant and  $T$  is the temperature. The time unit  $\tau$  is defined as  $(mr_C^2/\epsilon)^{1/2}$ . We used the modified velocity-Verlet algorithm with  $\lambda = 0.65$  to integrate the equation of Newton's motion.<sup>38</sup> To avoid

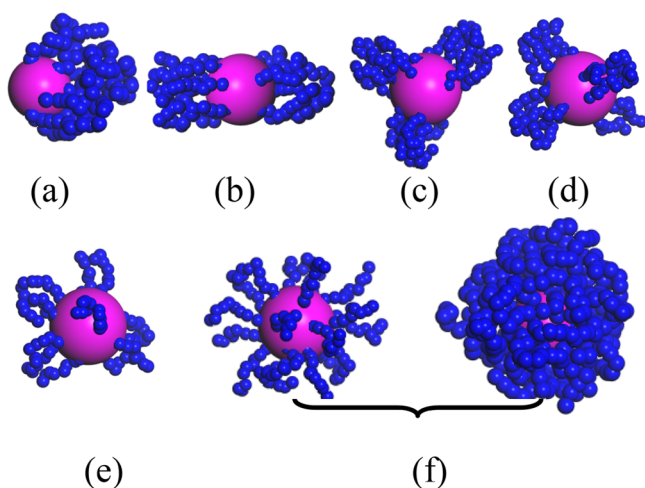
divergence of the simulation, the time step and amplitude of random noise are set as  $\Delta t = 0.03$  and  $\sigma = 3.0$ , respectively. The interaction parameters are chosen following the symmetric matrix below.

$$a_{ij} = \begin{pmatrix} & \text{H} & \text{T} & \text{S} \\ \text{H} & 25 & 25 & 25 \\ \text{T} & 25 & 25 & a_{\text{TS}} \\ \text{S} & 25 & a_{\text{TS}} & 25 \end{pmatrix} \quad (8)$$

Typically, the pairwise repulsive interaction parameters between the same type of DPD particles are set at  $a_{ii} = 25$ , according to the relationship between  $a_{ij}$  and Flory–Huggins interaction parameter  $\chi_{ij}$  at  $\rho = 3$ :  $a_{ij} \approx a_{ii} + 3.497\chi_{ij}$ . As  $a_{ij}$  increasing from 25, the incompatibility between  $i$  and  $j$  increases. As the nanoparticle is hydrophilic, the interaction parameters between nanoparticle and solvent is set as 25. In conditions mentioned above, each simulation needs at least  $1.0 \times 10^6$  steps for equilibrium, and we carried out  $2.0 \times 10^6$  steps for simulations.

## RESULTS AND DISCUSSION

In this work, we explored the influences of the interaction parameter between polymer chains tethered to nanoparticles and solvent ( $a_{\text{TS}}$ ), the size of nanoparticle  $D_{\text{H}}$ , and the number  $n$  and the length  $N$  of polymers on the surface patterning of nanoparticle. By changing the parameters mentioned above, diverse morphologies were observed, mainly consisting of five kinds of patchy structures and core–shell structure as shown in Figure 2. Patchy structures are the term for morphologies in



**Figure 2.** Representative morphologies of  $T_n\text{--H}$ : (a) monopatch, (b) dual-patch, (c) tripatch, (d) tetra-patch, (e) multipatch, (f) core–shell.

which all the chains aggregate and form well-defined patches. According to the number of patches  $N_{\text{patch}}$ , patchy structures are classified as monopatch ( $N_{\text{patch}} = 1$ ), dual-patch ( $N_{\text{patch}} = 2$ ), tripatch ( $N_{\text{patch}} = 3$ ), tetra-patch ( $N_{\text{patch}} = 4$ ), and multipatch ( $N_{\text{patch}} \geq 5$ ). While all the chains disperse uniformly or cover on the whole surface of nanoparticle, the morphology is denoted as core–shell structure.

**Influences of the Interaction Parameter and the Polymer Chain Length.** When the solvent changes from a good one to a poor one, the nanoparticle with polymer chains

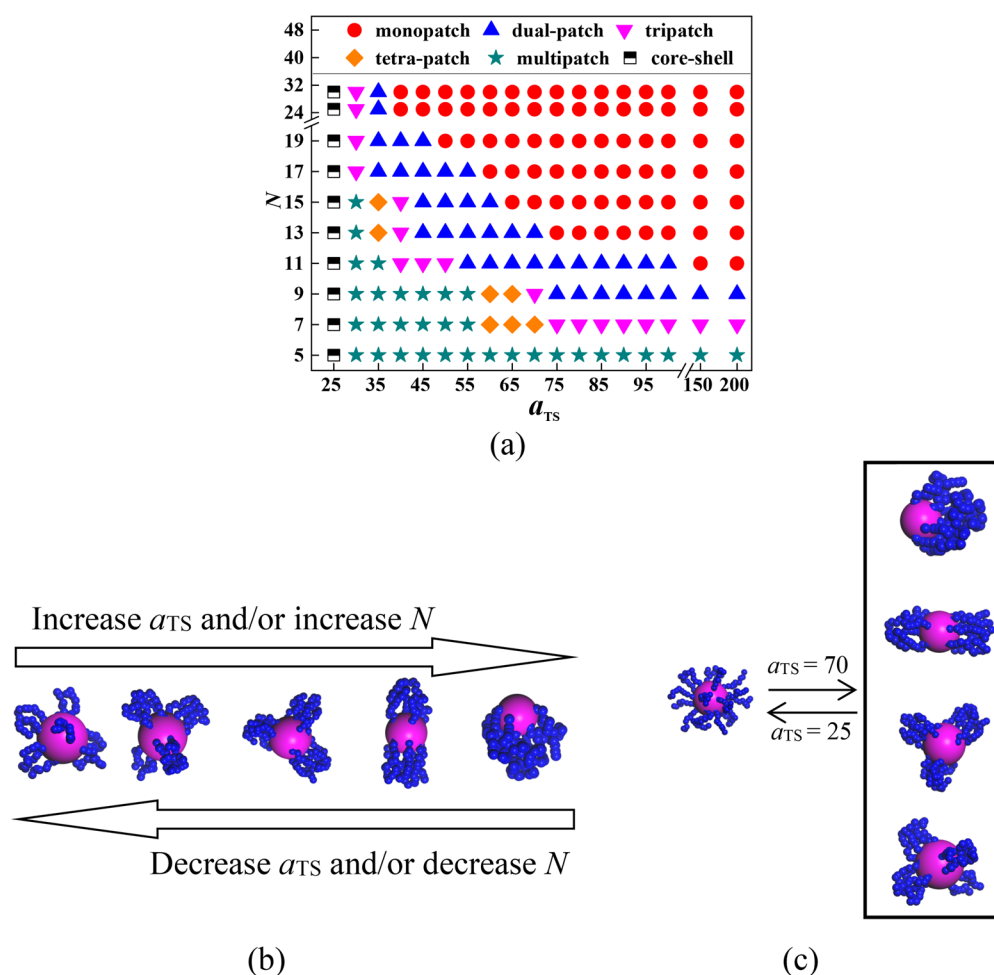
uniformly distributed would break into several patches.<sup>2,24,29,39</sup>

So we explored phase diagram of  $T_n\text{--H}$  ( $n = 15$ ,  $D_{\text{H}} = 6$ ) depending on the interaction parameter between polymer chains and solvents  $a_{\text{TS}}$  and the polymer chain length  $N$  (Figure 3a). From the vertical perspective, when  $a_{\text{TS}}$  is fixed at 25 where the solvent is good for polymers, the polymer chains disperse uniformly on the surface of nanoparticle, thus core–shell structure forms. When  $a_{\text{TS}}$  is changed from 30 to 200, the transitions from multipatch to tetra-patch or tripatch then to dual-patch to monopatch structure are observed as the chain length increases. It is reasonable that longer polymer chains are liable to aggregate. Horizontally, when  $N = 5$ ,  $T_{15}\text{--H}$  model mainly forms multipatch structure with  $a_{\text{TS}}$  increasing from 30 to 200, because too short polymer chains have weak interactions between each other. When  $N$  is equal to 7 or larger, the transition from core–shell structure to different patchy structures is observed as the interaction parameter  $a_{\text{TS}}$  increases. When the length of polymer is fixed, the volume fraction is determined at the same time. As the quality of solvent goes poorer, the number of patches decreases gradually. Less patches lead to lower surface free energy. Longer chains with stronger interaction (or in a poor solvent) result in less and bigger patches (Figure 3b). In experiment, the transition from core–shell in a good solvent to patchy particle in a poor solvent occurs by adding water to the nanoparticle solution in (dimethylformamide) DMF and the patch formation is reversible: upon dilution of the solution with DMF to a water concentration of  $C_w < 1$  vol %, the core–shell nanosphere morphology is recovered.<sup>29</sup> Our simulation results verified the reversibility of the formation of patchy particles: core–shell ( $a_{\text{TS}} = 25$ , good solvent)  $\rightarrow$  patchy particle ( $a_{\text{TS}} = 70$ , poor solvent)  $\rightarrow$  core–shell ( $a_{\text{TS}} = 25$ , good solvent) by tuning the interaction parameter between polymer chains tethered to nanoparticles and solvent which is a good index to change the solvent quality (Figure 3c). Our results are in good agreement with those of Hao et al.<sup>40</sup>

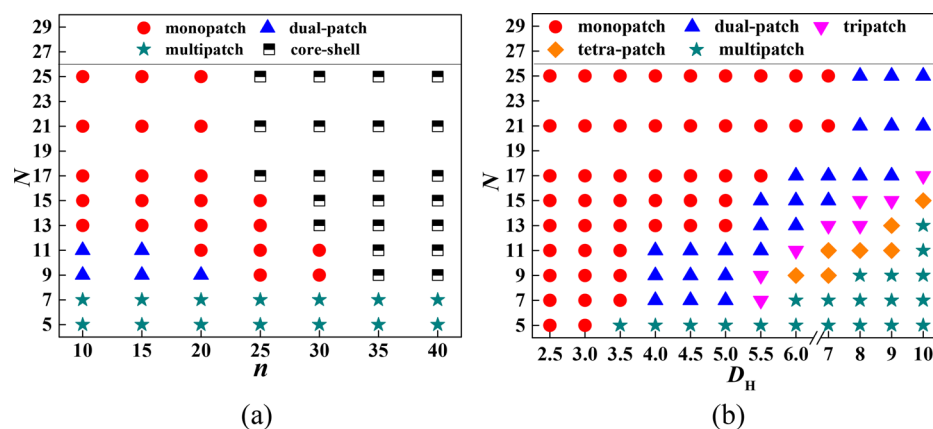
**Influences of the Number of Polymer Chains and the Size of Nanoparticle.** Figure 4a shows the dependency relationship between polymer chain length  $N$  and grafted polymer number  $n$  for model  $T_n\text{--H}$  ( $D_{\text{H}} = 4$ ,  $a_{\text{TS}} = 50$ ). From a vertical point of view, when  $n < 25$ , phase transition from multipatch to monopatch occurs as the chain length  $N$  increases. When  $25 \leq n \leq 30$ , we find the transition toward core–shell structure at last. When  $n > 30$ , the high grafting density makes polymers cover on the surface of nanoparticle, forming core–shell nanospheres. Horizontally, when the chain length  $N$  is fixed at 5 or 7, it forms multipatch structure because shorter chains cannot maintain uniform distribution. However, phase transitions from dual-patch to monopatch and monopatch to core–shell take place for  $N \geq 9$  as longer chains crush on the surface due to their stronger interaction. We conclude that, with the grafting density increasing, the probability for polymer chains to aggregate goes higher, and longer chains are helpful for the formation of patchy structure with less patches.

Figure 4b shows the influence of the nanoparticle diameter  $D_{\text{H}}$  and the chain length  $N$  with  $n = 15$  and  $a_{\text{TS}} = 50$ . As  $D_{\text{H}} < 3.5$ , only monopatch structure is observed. When  $D_{\text{H}} \geq 3.5$ , transition from multipatch to less patchy structure occurs as the chain length increases. However, the inverse transition from monopatch to dual-patch or multipatch takes place as the nanoparticle size increases with fixed polymer chain length. Compared with Figure 4a, the diameter of nanoparticle has





**Figure 3.** (a) Phase diagram of model polymer  $T_{15}$ -H depending on the interaction parameter  $a_{TS}$  and the polymer length  $N$  (the number of polymers tether to the nanoparticle  $n = 15$ , nanoparticle diameter  $D_H = 6.0$ ); (b) Schematic of the transition between different patchy structures; (c) The reversible transition between core-shell (in a good solvent) and patchy structure (in a poor solvent).

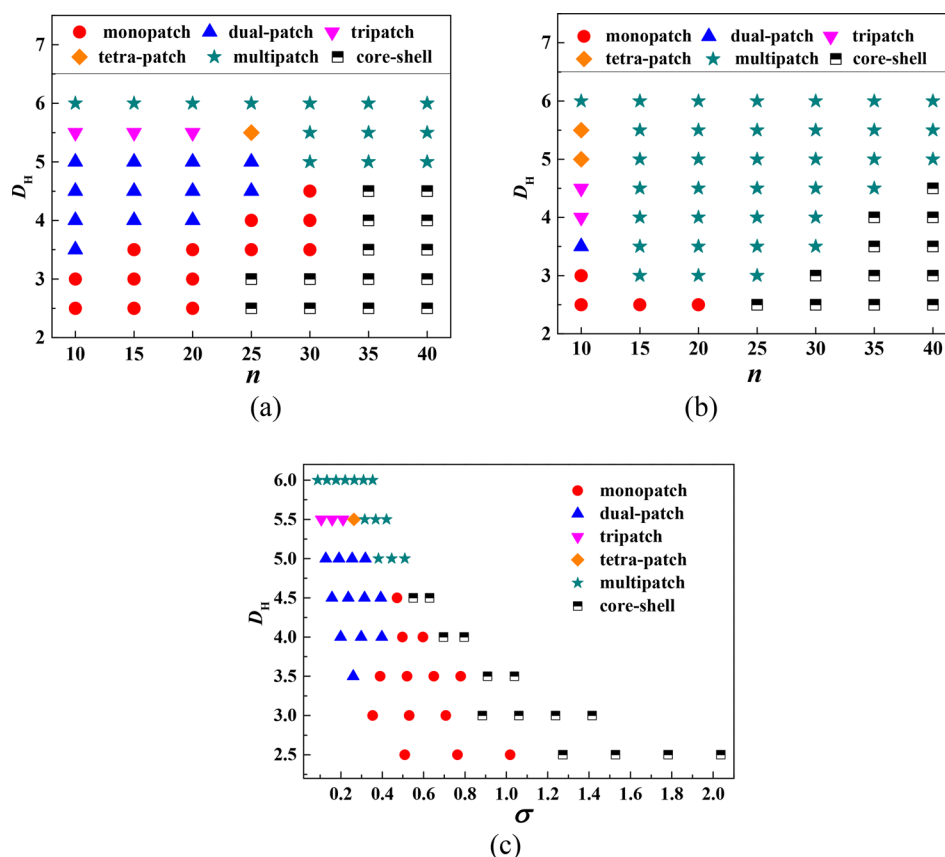


**Figure 4.** (a) Phase diagram of model polymer  $T_n$ -H, as a function of the polymer number  $n$  and polymer length  $N$  ( $D_H = 4$ ,  $a_{TS} = 50$ ); (b) Phase diagram of model polymer  $T_n$ -H, as a function of diameter of nanoparticle  $D_H$  and polymer length  $N$  ( $n = 15$ ,  $a_{TS} = 50$ ).

stronger influence on the morphology. Thus, adjusting the diameter of nanoparticle can be used as a powerful method to prepare well-defined patchy structure which has been used in experiments.<sup>29</sup>

**The Dependency Relationship between the Number of Polymer Chains and the Size of Nanoparticle.** It is obviously shown in Figure 4b that when  $n = 15$  and  $N = 9$ ,  $T_n$ -

H has rich morphologies. So we explored the dependency relationship between the number of tethered chains to the nanoparticle and diameter of nanoparticle at  $N = 9$  and  $a_{TS} = 50$  (Figure 5a). When  $n$  is less than 20, increasing diameter leads to the number of patches increase from 1 to 3. As  $n$  equals to 25 or 30, and the diameter  $D_H = 2.5$  and 3.0, the core-shell structure forms as the high grafting density



**Figure 5.** (a) Phase diagram of model polymer  $T_n-H$ , as a function of number of tethered polymers  $n$  and diameter of nanoparticle ( $N = 9$ ,  $a_{TS} = 50$ ); (b) Phase diagram of model polymer  $T_n-H$ , as a function of number of tethered polymers  $n$ , and diameter of nanoparticle ( $N = 7$ ,  $a_{TS} = 50$ ); (c) Phase diagram of model polymer  $T_n-H$  as a function of the nanoparticle diameter  $D_H$  and grafting density  $\sigma$  (All data points are corresponding to that of panel a in the condition  $N = 9$ ,  $a_{TS} = 50$ ,  $n = 10-40$  and  $D_H = 2.5-6$ ).

polymers cover on the surface of small nanoparticle. While  $n$  is fixed at 25 and  $D_H$  increases from 3.5 to 6, the distances between polymers are so far that it is hard for them to get together, the number of patches increasing from 1 to 4. When  $D_H > 3$  and  $n = 30$ , increasing  $D_H$  makes the morphology transform from monopatch to multipatch. The multipatch forms because the high grafting density offsets partly the influence brought about by bigger size of nanoparticle. When  $n \geq 35$ , the morphology transforms from core-shell to multipatch with the nanoparticle size increasing. The larger the nanoparticle size is, the larger the distances between polymers are, thus it is more beneficial to form the patchy structure.

Figure 5b presents the phase diagram at  $N = 7$  and  $a_{TS} = 50$ . Most of the morphologies are multipatch and core-shell. However, as  $n = 10$ , polymers are far from each other, and well-defined patchy structure occurs with diameter of nanoparticle increasing. We conclude that the number of patches  $N_{patch}$  is controlled by diameter of nanoparticle when  $T_n-H$  has lower grafting density. For  $D_H = 2.5$ , with the grafting density increasing, the morphology transforms from monopatch to core-shell structure. Higher grafting density strengthens the interaction between polymers, so they distribute uniformly. By comparing Figure 5a and b, when  $T_n-H$  has proper grafting density, the length of polymer is the main factor that affects the morphology.

To compare with the experimental results, here we introduce the grafting density ( $\sigma$ , in unit of chains/ $r_c^2$ ) representing the

ratio of the number of polymer chains to the surface area of nanoparticle:

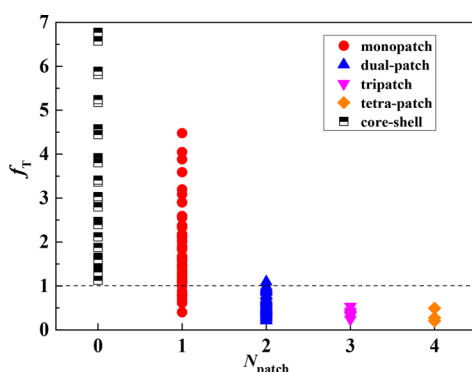
$$\sigma = \frac{n}{\pi D_H^2}$$

Figure 5c presents the phase diagram as a function of the nanoparticle diameter  $D_H$  and grafting density  $\sigma$  where all data points are corresponding to that of Figure 5a. We can clearly see that when the grafting density is fixed, the increasing of  $D_H$  will lead to different phase transition paths (from monopatch to dual-patch to multipatch, from monopatch to core-shell) which matches well with the experimental results,<sup>29</sup> except the boundary of patch and core-shell is a little different.

**Analysis of the Effective Volume Fraction of Polymers vs. the Number of Patches on the Nanoparticle.** Here we define the effective volume fraction  $f_T$  of polymers as follows

$$f_T = \frac{nN^\nu r_c^3}{D_H^3}$$

where  $n$ ,  $N$ ,  $D_H$ , and  $r_c$  represent the number of polymers grafted onto one nanoparticle, the polymers length, nanoparticle diameter and the length unit ( $r_c = 1$ ), respectively. The coefficient ( $\nu$ ) of  $N$  is obtained from the corresponding relationship root-mean-square end-to-end distance of polymers  $R_e \sim N^\nu$ . We denoted the  $N_{patch}$  of core-shell structure is 0. All data points of the diagram of  $f_T - N_{patch}$  shown in Figure 6 are in the condition  $a_{TS} = 50$ ,  $N = 7, 9, 11$ , and  $13$ ,  $n = 10-40$  and



**Figure 6.** Relationship between the effective volume fraction of polymers  $f_T$  and the number of patches  $N_{\text{patch}}$ . All data points are in the condition,  $a_{TS} = 50$ ,  $N = 7, 9, 11$ , and  $13$ ,  $n = 10\text{--}40$  and  $D_H = 2.5\text{--}6$ .

$D_H = 2.5\text{--}6$ . The multipatch points are not shown here as its classification is complex. By and large, with increasing  $f_T$ , the phase transition complies with the path: tetra-patch  $\rightarrow$  tripatch  $\rightarrow$  dual-patch  $\rightarrow$  monopatch  $\rightarrow$  core-shell. So we can use  $f_T$  as an indicator to guide the preparation of well-defined patchy structure. That is to say, when  $f_T$  is lower than 1, patch structures ( $N_{\text{patch}} = 2\text{--}4$ ) are easily to be obtained. When it is larger than 1, monopatch or core-shell structure forms.

## CONCLUSIONS

We have adopted DPD to study the morphology of nanoparticle with several tethered polymer chains in good or poor solvents based on the experimental research (ref 29). By changing the interaction parameter between polymer chains tethered to nanoparticle and solvent, the size of nanoparticle, and the number and the length  $N$  of polymers on the surface patterning of nanoparticle, various patchy structures with different numbers of patches ( $N_{\text{patch}}$ ) and core-shell structure were observed. Increasing chain length and interaction strength between polymers and solvent are beneficial to transform from core-shell to patchy structures. The reversible transition between core-shell (in a good solvent) and patchy structure (in a poor solvent) in experiment is verified in the simulation. Longer chains with stronger interaction (or in a poor solvent) result in less and bigger patches, saying,  $N_{\text{patch}}$  decreases as chain length and interaction strength increase. Increasing nanoparticle diameter, the inverse phase transition path ( $N_{\text{patch}}$  increases) occurs. In addition, increasing chain number at long chain length or smaller nanoparticle size is helpful to transform from monopatch to core-shell structure. We introduce the effective volume fraction  $f_T$  as an indicator to guide the preparation of well-defined patchy structure. When  $f_T < 1$ , patch structures ( $N_{\text{patch}} = 2\text{--}4$ ) are easily to be obtained. When  $f_T > 1$ , monopatch or core-shell structure forms. The phase diagram for the nanoparticle diameter  $D_H$  vs. grafting density  $\sigma$  matches well with the experimental results. Our results will guide researchers to prepare the colloids with well-defined patches. The anisotropic patchy particles can self-assemble into elaborate superstructures, which are potential blocking materials for drug delivery, sensors, and electronics.

## AUTHOR INFORMATION

### Corresponding Author

\*E-mail: wangrong@nju.edu.cn.

### ORCID

Qiuyan Yu: 0000-0002-3792-2514

Rong Wang: 0000-0001-7525-1400

### Author Contributions

<sup>†</sup>These authors contribute equally.

### Notes

The authors declare no competing financial interest.

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